Anders Garberg

Engineer, Robotics & Autonomy (B.Sc.) • M.Sc. Student, Cybernetics & Robotics, NTNU

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PROFILE

Robotics and autonomy engineer and M.Sc. student in Cybernetics and Robotics at NTNU. Built ROS 2 software and an NVIDIA Isaac Sim test environment for KONGSBERG's Lone Wolf unmanned ground vehicle. Currently works on guidance and navigation for an autonomous surface vessel in Marinor NTNU. Main interests are autonomous systems, maritime robotics and robotics software.

RELEVANT TECHNOLOGIES

Languages & Frameworks: C++; Python; Qt; MATLAB; PLC Structured Text; TypeScript/React

Robotics & Autonomy: ROS 2; NVIDIA Isaac Sim; URDF/USD; guidance and navigation; sensor integration; unmanned ground vehicles; autonomous surface vessels

Control & Modelling: control theory; MPC; system identification; Simulink; OPC

Systems: Linux; embedded Linux (NVIDIA Jetson); telemetry and video streaming; Docker Compose; Git; networking

RELEVANT EXPERIENCE

KONGSBERG
Jun 2026 – Aug 2026

Summer Intern - Software, Lone Wolf — Kongsberg, Norway

- Part of an eight-person student team that developed sensor integration and a control system for self-driving and object tracking on an unmanned ground vehicle.
- Developed the ICS operator panel in Qt and a gateway that streams ROS 2 topics from the vehicle over the network to the panel.
- Set up launch scripts on the vehicle's NVIDIA Jetson that start all ROS 2 nodes, the telemetry gateway and the video stream.
- Developed, tested and verified ROS 2 software in C++ on the physical vehicle.

Marinor NTNU
Sep 2026 – Present

Autonomy Engineer (Voluntary technical position) — Trondheim, Norway

- Develops guidance and navigation functionality for an autonomous surface vessel in a multidisciplinary student team.
- Marinor builds autonomous maritime vessels and is a member of the Ocean Autonomy Cluster.

SELECTED PROJECTS

B.Sc. Thesis: Lone Wolf Synthetic Test Environment (KONGSBERG, grade A)
Jan 2026 – Jun 2026

- Developed a synthetic test environment in NVIDIA Isaac Sim for KONGSBERG's Lone Wolf UGV, enabling repeatable testing of autonomy software without access to the physical robot. Team of four.
- Responsible for integrating our ROS 2 software with Isaac Sim, including correct import of the robot's URDF and USD models and setup of the ROS 2 graphs in the simulator.
- Validated against physical recordings with comparable odometry response and consistent depth camera data. Published in NVA: hdl.handle.net/11250/5548753

Automation Project - Water Tank Control
2025

- Developed a discrete-time PID/PI-D controller in PLC Structured Text, including tracking/bumpless transfer, anti-windup, derivative filtering and lead-lag feedforward for disturbance compensation.
- Built a closed-loop simulation setup by coupling PLC and Simulink via OPC, plus data-driven modelling (MATLAB System Identification / NLARX) to tune and validate behaviour before deployment.

Informize - digital information screens (web app)
2025 – Present

- Full-stack web app (TypeScript/React) for running unattended information screens around the clock. Developed both with and without AI coding assistants.

OTHER EXPERIENCE

NTNU

Teaching Assistant • Trondheim, Norway

Guides students through problem sets and lab work, gives academic follow-up and supports course staff in delivery.

TDT4136 Introduction to Artificial Intelligence — Aug 2026 – Present
IELET2107 Robotics & Machine Vision — Aug 2026 – Present
AIST2102 AI for Robotics & Automation — Jan 2026 – May 2026
ISTT1002 Statistics — Aug 2025 – Dec 2025, Aug 2026 – Present

Trondheim Kino

Jun 2021 – Present

Shift Leader (part-time) — Trondheim, Norway

- Responsible for daily operations, staff training and customer service. Also responsible for the technical side of screenings, including projectors, playback servers and sound systems.

Norwegian Armed Forces

Aug 2022 – Jan 2024

Communications Operator — Ground Based Air Defense — Ørland, Norway

- Operational experience with communications systems in a high-readiness environment as part of the NATO Response Force; ensured secure and effective communication between GBAD systems under demanding conditions.

EDUCATION

NTNU

Trondheim, Norway

M.Sc. in Cybernetics and Robotics — Aug 2026 – Jun 2028

- Specialization in Robotics and Vehicle Control.

B.Sc. in Automation and Intelligent Systems — Aug 2023 – Jun 2026

- Three-year degree in control engineering, robotics, autonomy and intelligent systems.
- Grade average 4.5/5 (A). Thesis grade A.

LEADERSHIP & VOLUNTEERING

Elektra Student Association

Student association for electrical engineering students at NTNU • Trondheim, Norway

Auditor — Aug 2026 – Present

Financial Manager / Treasurer — Oct 2024 – Nov 2025

- Owned budgeting, accounting and financial planning for an association with several hundred members.

Event Committee Member, incl. Financial Manager — Aug 2023 – Jun 2026

UKA

Aug 2023 – Nov 2023

Bartender (Voluntary) — Trondheim, Norway

Stormrevyen

Jun 2021 – May 2022

Manager and Treasurer (Voluntary) — Trondheim, Norway

ADDITIONAL

Languages: Norwegian (native), English (fluent) | **Citizenship**: Norwegian | **Driver's license**: Category B

Security clearance: Held during service with the Norwegian Armed Forces (2022–2024)

Availability: Summer internship 2027; part-time from Jan 2027; full-time from Jun 2028